

Recall For v-space V with basis $B = (v_1, \dots, v_n)$ and linear map $f: V \rightarrow W$
 W with basis $C = (w_1, \dots, w_k)$

Having bases
gives a dictionary!

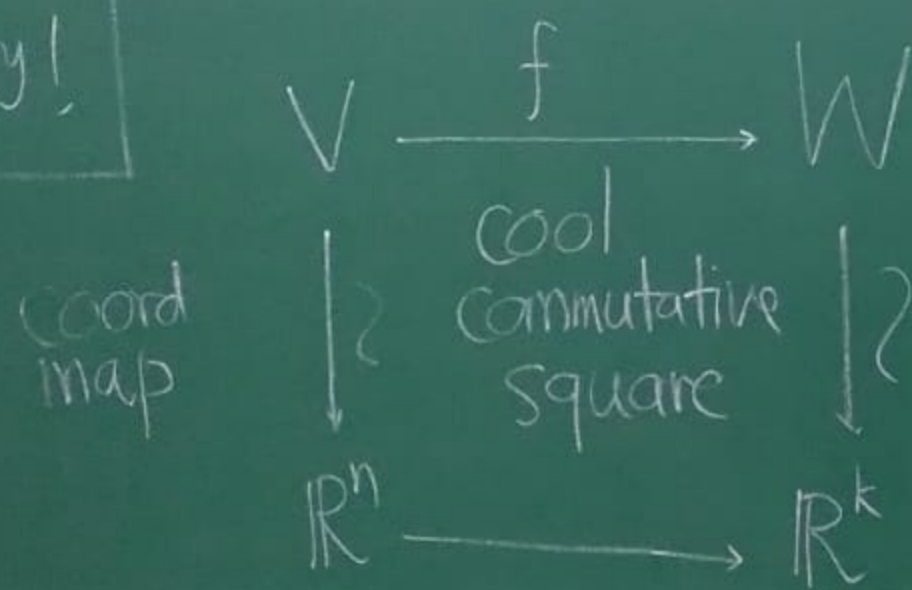


Diagram illustrating the relationship between vector spaces and their coordinate representations:

$$\begin{array}{ccc}
 v & \xrightarrow{\text{lin map}} & f(v) \\
 \downarrow & & \downarrow \\
 [v]_B & \xrightarrow{\text{matrix mult}} & [f(v)]_C = M_{B,C}(f) [v]_B
 \end{array}$$

Change of basis

① V has ordered bases

$B = (v_1, \dots, v_n)$ and $B' = (v'_1, \dots, v'_n)$

Change of basis matrix P

$$\left. \begin{array}{l} v'_1 = p_{11}v_1 + \dots + p_{n1}v_n \\ \vdots \\ v'_n = p_{1n}v_1 + \dots + p_{nn}v_n \end{array} \right\} [v'_1 \dots v'_n] = [v_1 \dots v_n] P$$
$$\begin{array}{l} B' = BP \\ B = B' P^{-1} \end{array}$$

Given $\left\{ \begin{array}{l} V \xrightarrow{f} W \\ \text{linear} \end{array} \right\} \left\{ \begin{array}{l} C = (w_1, \dots, w_k) \\ C' = (w'_1, \dots, w'_k) \end{array} \right\} \begin{array}{l} \text{ordered} \\ \text{bases} \\ \text{for } W \end{array}$

Change of basis for linear map f

let $C' = CQ$ Q : change of basis in W

let $M_{B,C}(f) = A$ i.e. $[f(v)]_C = A[v]_B$

Want: $[f(v)]_{C'} = ?? [v]_{B'}$ Ans: ?? = $Q^{-1}AP$

$$[f(v)]_{C'} = Q^{-1} [f(v)]_C = Q^{-1} A [v]_B = Q^{-1} A P [v]_{B'}$$

Note: If $V=W$, we call such a linear map $V \xrightarrow{T} V$ an operator on V . In this case, we want to use the same basis for domain & codomain. Change of basis formula becomes $P^{-1}AP$.

QAP draws many threads together!

- row and column operations
- rank nullity thm (concrete and abstract)
- change of basis

→ Articulate the whole picture for yourself. Eg. →

Matrix of f using bases
obtained from pf of
rank-nullity thm

$V = v_1, \dots, v_r, k_1, \dots, k_n$

$N: f(v_1), \dots, f(v_r)$, extend in any way

$$\begin{array}{c} f(v_1) \\ \vdots \\ f(v_r) \\ * \\ * \\ \vdots \end{array} \begin{bmatrix} v_1 & v_2 & \dots & v_r & k_1 & \dots & k_n \\ \hline & I_{r \times r} & & & 0 & & \\ & & & & 0 & & 0 \end{bmatrix}$$

Here Endeth^{*} Linear Algebra O.

(^{*} sort of, at least for now.)

Thm: There is a unique function $(n \times n \text{ matrices}) \rightarrow \mathbb{R}$
that satisfies the following properties, when regarded
as a function from n ordered col vectors

- ① $\det$ is a multilinear function i.e. linear in col $\# i$ if all other cols are fixed
- ② $\det$ is "alternating" i.e. $\det(A) = 0$ if any two consecutive cols are equal.
- ③ $\det(I) = 1$

[Note: It will turn out that $\begin{cases} \text{"column" can be replaced everywhere by "row" and} \\ \text{"consecutive" can be dropped} \end{cases}$]

Remarks • If we want $\det(c_1, c_2, c_3)$ to behave like our intuitive idea of volume of parallelopiped formed by vectors c_1, c_2, c_3 then properties ①, ②, ③ make geometric sense! ②, ③ are easy. Property ① says, e.g.

$$\det(c_1, c_2, c_3) + \det(c'_1, c_2, c_3) = \det(c_1 + c'_1, c_2, c_3). \text{ See why we would want this!}$$

• Let's see now ①, ②, ③ force the formula for $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Note $\text{col } 1 = ae_1 + ce_2$, $\text{col } 2 = be_1 + de_2$

$$\det(ae_1 + be_2, ce_1 + de_2) = \det(ae_1, ce_1 + de_2) + \det(ce_2, be_1 + de_2) \quad \left\{ \begin{array}{l} \text{using linearity in first column} \end{array} \right.$$

$$= a \det(e_1, ce_1 + de_2) + c \det(e_2, be_1 + de_2)$$

$$= ac \underbrace{\det(e_1, e_1)}_{\text{repeated col}} + ad \underbrace{\det(e_1, e_2)}_I + cb \underbrace{\det(e_2, e_1)}_{II} + cd \underbrace{\det(e_2, e_2)}_{\text{repeated col}} \quad \left\{ \begin{array}{l} \text{using linearity} \\ \text{in 2nd column} \end{array} \right.$$

$$= ad - bc \quad (\text{See left corner.}) - \det(e_1, e_2)$$

Use $\det(e_1 + e_2, e_1 + e_2) = 0$
(by ②) with multilinearity
and ② to see that
 $\det(e_1, e_2) + \det(e_2, e_1) = 0$